Name and Surname:	
Subject:	Mathematics
Grade:	11AG
Duration:	40 minutes
Exam:	Final Exam
Name of the Teacher:	Anvarbek Khaydarov
Date:	26/10/2026
Score:	_____/100%

INSTRUCTIONS

1. Read all questions carefully before answering.
2. Use a blue pen or pencil to answer all questions.
3. Write your answer to each question in the space provided.
4. No electronic devices (phones, smartwatches, headphones) are allowed.
5. Do not leave the class/exam hall in the first 30 minutes

MATHEMATICS — FINAL EXAM

Grade 11AG · 26/10/2026 · Duration: 40 minutes · Total: 100 marks

Answer all questions in the space provided. Show your working — method carries marks. Angles are in radians throughout.

Section A: 8 questions × 3 marks = 24. Section B: 8 questions × 5 marks = 40. Section C: 4 questions × 9 marks = 36. Easy 40% · Medium 40% · Hard 20%.

SECTION A — Short answers (8 questions × 3 marks = 24 marks)

1. Find $\lim_{x \to 2}(3x^2 - 5)$. [3 marks]

2. Find $\lim_{x \to 3} (x^2 - 9)/(x - 3)$. [3 marks]

3. Find $\lim_{x \to \infty} 1/x^2$. [3 marks]

4. Find $\lim_{x \to \infty} (5x^2 + 1)/(2x^2 - x)$. [3 marks]

5. Differentiate $y = 4x^5$. [3 marks]

6. Differentiate $y = 3x^2 - 7x + 2$. [3 marks]

7. Differentiate $y = \sin x + e^x$. [3 marks]

8. Differentiate $y = \ln x - \cos x$. [3 marks]

SECTION B — Show your working (8 questions × 5 marks = 40 marks)

9. Find $\lim_{x \to 1} (x^2 + 2x - 3)/(x - 1)$. [5 marks]

10. Find $\lim_{x \to 0} (\sqrt{(x + 4)} - 2)/x$. [5 marks]

11. Find $\lim_{x \to \infty} (3x^3 - 2x)/(6x^3 + 5)$. [5 marks]

12. Differentiate $y = (2x + 1)^5$. [5 marks]

13. Differentiate $y = x^2e^x$, giving your answer in factorised form. [5 marks]

14. Differentiate $y = (\sin x)/x$. [5 marks]

15. Differentiate $y = \ln(3x^2 + 1)$. [5 marks]

16. Differentiate $y = \tan 4x$, and find the gradient of the curve at $x = 0$. [5 marks]

SECTION C — Extended questions (4 questions × 9 marks = 36 marks)

17. Evaluate the following limits:

(a) Find $\lim_{x\to 2} (x^3 - 8)/(x^2 - 4)$. [3 marks]

(b) Find $\lim_{x\to\infty} (\sqrt{(x^2 + 3x)} - x)$. [3 marks]

(c) Explain why $\lim_{x\to 0} |x|/x$ does not exist. [3 marks]

18. Differentiate, simplifying your answer where possible:

(a) $y = \sqrt{(x^2 + 9)}$ [3 marks]

(b) $y = e^{\sin x}$ [3 marks]

(c) $y = \ln(\cos x)$ [3 marks]

19. A curve has equation $y = x^2 \sin x$.

(a) Find dy/dx. [3 marks]

(b) Find d^2y/dx^2. [4 marks]

(c) Find the value of dy/dx at $x = 0$. [2 marks]

20. A curve has equation $y = (\ln x)/x^2$, for $x > 0$.

(a) Find dy/dx. [4 marks]

(b) Show that $dy/dx = 0$ when $x = \sqrt{e}$. [3 marks]

(c) State, with a reason, the value of $\lim_{x\to\infty} (\ln x)/x^2$. [2 marks]